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Recently Discovered Mémoire Written by the Forsters in the Light of Other Works of Their Authorship: Comparative and Navigational Reflections

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Resumen

Un informe recién descubierto, escrito por los Forster a la luz de otros trabajos de su autoría: reflexiones comparativas y navegacionales

La originalidad del presente artículo reside en la comparación del recién descubierto manuscrito titulado Mémoire Sur Waïhou, ou L'Isle de Pâques, escrito en francés por Johann Reinhold Forster y George Forster, con obras suyas bien conocidas: A Voyage, Resolution Journals y Observations. Mémoire –conservado en la Biblioteca Jaguelónica en Cracovia (Polonia) –constituye el núcleo de nuestro trabajo; se discuten su forma y contenido, su mensaje y objetivos, así como las diferencias respecto a otras obras Forsterianas. Sobre todo, se descubren motivos de carácter novedoso, ausentes en otros textos de los Forster, pero incluidos en el manuscrito. Estos elementos arrojan una nueva luz sobre las investigaciones acerca de Isla de Pascua. Además, ponemos especial énfasis en cuestiones de geografía y navegación, hasta ahora poco estudiados en el contexto de Isla de Pascua. Otro asunto no menos importante que tematizamos, es nuestro intento de establecer la verdadera autoría del documento.

Summary

The originality of the present paper lies in a comparative study between a recently discovered *Mémoire Sur Waïhou, ou L'Isle de Pâques*, originally written in French by Johann Reinhold Forster and George Forster, and their famous works: *A Voyage, Resolution Journals* and *Observations. Mémoire*, kept in the Jagiellonian Library in Cracow, is the core of the comparative work we are seeking to conduct. The document's form and content are discussed, as well as its message, aims, and the differences with other Forsterian works – especially new remarks and commentaries, absent in the known publications by the Forsters. These motifs shed new light on the research on Easter Island. Moreover, a special emphasis is put on such questions as geography and navigation, seldom studied in the context of Rapa Nui. Another important issue is determination of the true author of *Mémoire*.

Introduction: A Short Biography of the Forsters and *Mémoire* Presentation

Johann Reinhold Forster and his son George were born near Gdańsk in the Polish-Lithuanian Commonwealth in 1729 and 1754 respectively. Their progenitor came indirectly from Scotland. As a result of intermarriages in the area relatively close to Gdańsk, in their genetic constitution they were also Polish and German.

It is a well-known fact that they sailed as supernumeraries on board the *Resolution* during Cook's Second Voyage (1772–1775) and then became two of the most accomplished European writers on the 18th century natural science and Pacific exploration.

However, the turning point in their life came in 1765, when Johann – then a pastor in Mokry Dwór – was invited to Russia to the Volga region, to carry out the research on German settlements. He went there with his son George. Having spent about a

year in Russia, they sailed directly to London where they disembarked from the packet *Liebste* in 1766.

The papers on natural history research on the Volga region made J. R. Forster known among the natural science community in England and contributed towards his membership in the Royal Society[1]. This fact, and his subsequent works, increased his fame in England as an excellent natural scientist.

In London J. R. Forster made the acquaintance of Joseph Banks and Daniel Charles Solander, participants in Cook's First Voyage. Banks, however, withdrew from the participation in the captain's Second Voyage. This indirectly led to King George III approving the appointment of Johann Forster as a principal naturalist and George Forster, a talented draughtsman, as his assistant on board the *Resolution*.

By the time the Second Expedition returned, the Forsters had collected thousands of natural history specimens, together with illustrations, and notes on ethnology as well as Pacific region languages. In his *Journal* Johann made highly detailed daily entries. He left Plymouth on the Voyage on the understanding that the right to publish his *Journal* and scientific results was verbally given to him by British officials. However, Johann was prohibited to publish his memoir. Instead he published *Observations* and George issued *A Voyage*. The latter and its German translation titled *Reise um die Welt*[2] made George rather famous on the continent, especially in Germany. However, it did not bring the Forsters recognition in England or the eagerly awaited financial support and stabilization in Britain. Johann was put to prison due to his debts.

George's visit to Germany secured for his father an appointment in 1779 as professor at the University of Halle. Johann lectured there until shortly before his death in 1798.

In 1784, George went as a professor to the Vilna University. Later the Russians invited him to participate in their Pacific expedition. Nevertheless, the Russian plan failed. A Spanish expedition, also to the Pacific region, in which George might have taken part, also came to nothing.

In March 1793 George went to Paris to petition for the annexation of the occupied Rhenish territories to the French Republic. As a result, the German Emperor declared George a traitor. Therefore, the author of *A Voyage* faced the death penalty within the Emperor's territories. He deceased in Paris in his bed at the age of only 39, on 10th or 12th January 1794, in utter penury.

The main concern of ours in this paper, namely the *Mémoire Sur Waïhou, ou L'Isle de Pâques*, signed by Jean Renaud (*sic*) Forster – presently a part of the so-called Berlin Collection of the Jagiellonian Library in Cracow (Poland) – is a manuscript written in French and bound in ornamented leather. It consists of 26 large folios, enriched by carefully selected illustrations. The illustrations include two maps: *A Chart of the Southern Hemisphere* by George Forster and a map of Easter Island, as well as two famous portraits of Easter Islanders engraved after the drawings by William Hodges, and a previously unknown, anonymous watercolour depicting the island's landscape with the indigenous people, *moai* figures on an *ahu* and a local canoe[3]. For an extended manuscript description, we kindly refer the reader to Jakubowska (2014a, 2014b).

The *Mémoire* was not included in any Forsterian bibliography. Up to the year 2014 it was only briefly studied by Chrobak (2010), who participated in a project aimed at the investigation of French documents included in the aforementioned Berlin Collection (Rzepka *et al.* 2011); however, the results of her work remained virtually unknown in the circles of Easter Island researchers. The first approach to a detailed analysis and assessment of the content and style of the document, undertaken by Jakubowska (2014a, 2014b), revealed that the material authorship of Johann Reinhold Forster was doubtful; the ensuing graphological expertise confirmed the hypothesis that the handwriting of the *Mémoire* belongs rather to his son, George Forster (Bartoszyński 2014). Why should the naturalists conceal the true identity of the manuscript's material author? As Vorbrich (2009: 98) states, the Forsters used to adopt a 'peculiar policy' regarding the true identity of the author of their works, in the sense that they were not too keen to disclose the authorship in a clear manner. However, we shall elaborate on that topic when discussing the rationale behind composing the *Mémoire*.

[1]After Cook's Second Voyage in 1775, Johann Forster, in recognition of this work on natural history, was also elected a member of the Scientific Society in Gdańsk.

[2]We are quoting abbreviated titles here.

[3]We conjecture that it is quite similar to a sketch of a canoe by J. R. Forster (cf. Hoare [ed.] 1982: between 464–465), although much more elaborate; Coote suggests it may be a work by G. Forster (*apud* Jakubowska 2014b: XIV).

As far as the period of 18th century discovery voyages is concerned, researchers specialized in Easter Island studies had at their disposal mainly excerpts of logbooks, journals and memoirs written by captains and other sailors taking part in those enterprises. To some extent, Spanish documents of 1770 are exceptional, as for the crew under the command of Felipe González the exploration of Easter Island (or rather 'Isla de David' – *sic*) was a principal, albeit not a sole objective. In this context the *Mémoire* is a real treasure for us. It is a complete essay dealing with Rapa Nui as its unique subject, summarizing almost all the relevant knowledge obtained during James Cook's Second Circumnavigation (1772–1775). Moreover, it is composed with a clear philosophical (and political) idea in mind, and can also be regarded as an (early) anthropological treatise.

In the following sections of this present paper we shall deal with principal similarities and differences between the *Mémoire* and the most famous works by the Forsters, already mentioned, namely:

> *a) A Voyage round the World, in His Britannic Majesty's Sloop, Resolution, commanded by Capt. James Cook, during the Years 1772, 3, 4 and 5…*, by George Forster (1777),
>
> *b) The Resolution Journal of Johann Reinhold Forster 1772–1775* (Hoare [ed.] 1982),
>
> *c) Observations Made During the Voyage Round the World*, by Johann Reinhold Forster (1778).

We shall also discuss briefly the probable addressee of the manuscript – whose figure is closely related to the political reasons that incited the Forsters to write the essay. Last but not least, we would like to address important geographical and navigational problems relevant to our topic. As to the *Mémoire* itself, we are referring to and citing from the English translation of the document included in Jakubowska (2014b)[4].

The *Mémoire* and other Forsterian works: principal correspondences and analogies

What makes the whole matter particularly interesting is the fact that every work in question was written for different reasons and with different aims. The *Journal* by Johann Forster, published two centuries after the Voyage, was most probably issued without any editorial interference. However, the naturalist, while composing it, thought of himself as the principal or even sole author of the Voyage narrative to appear in print (*cf. e.g.* Vorbrich 2011: 187). Clearly he thought of subsequent reworking and polishing his notes taken 'live' during the circumnavigation; nevertheless, for reasons we shall elaborate on later, his work was not published during his lifetime.

What was published, however, was *A Voyage* by his son George: an extensive, two-volume account, with a great probability based on the *Journal*, but also on George's own experience, and showing a characteristic style as if of a novel, vivid, sometimes dramatic or poetic. Certainly, a unique and valuable work, unjustly criticized and accused of an alleged plagiarism (e.g. Wales 1778).

Prohibited from publishing his *Journal*, Johann Forster prepared another work: *Observations*, which deals with many experiences and reflexions originated during the Voyage, but ordered thematically and not chronologically. It can also be treated as J. R. Forster's philosophical manifesto.

Due to the nature of the first two publications – a journal written in the real time and a stylized chronicled narrative of the voyage, respectively – the information about Easter Island we are interested to extract is fragmented, scattered among various episodes and descriptions. This information is gradually revealed, and sometimes repeated, in accordance with the sequence of events occurring in the course of the visit on Rapa Nui.

The *Mémoire*, as to its content, organisation and style, constitutes a kind of fusion between the narrative works (*A Voyage* and the *Journal*) and *Observations*. Many elements that make up its content are clearly taken directly from the Voyage notes and experiences. We can cite such examples as:

Cook's crew condition and moods before reaching Easter Island (*apud* Jakubowska 2014b: 74–75; Forster 1777: 550, 552; Hoare [ed.] 1982: 460–461, 464),

Disappointment at reaching an island offering poor resources for exhausted sailors (*apud* Jakubowska 2014b: 76; Forster 1777: 552–553, 556; Hoare [ed.] 1982: 461, 463),

The first encounter with the indigenous people on shore (*apud* Jakubowska 2014b: 76–77; Forster 1777: 563; Hoare [ed.] 1982: 465–466),

[4]The publication also contains a reproduction and transcription of the original document.

First impressions on the character and nature of Rapanui people (hospitality, reaction to firearms, 'theft'/'cheating' etc.) (*apud* Jakubowska 2014b: 77; Forster 1777: e.g. 557, 563, 565–566, 571–572, 579–580, 582, 584–585, 590; Hoare [ed.] 1982: 467–468, 470–472, 474–476),

Encounter with a generous Rapanui couple, offering sweet potatoes and water to reconnaissance group members (*apud* Jakubowska 2014b: 81; Forster 1777: 588[5]; Hoare [ed.] 1982: 472),

Encounter with the alleged 'king' of Easter Island (*apud* Jakubowska 2014b: 81; Forster 1777: 588–589[6]; Hoare [ed.] 1982: 471–472),

Encounter with a Rapanui man and his son who helped the tired J. R. Forster to reach the ship (*apud* Jakubowska 2014b: 82; Forster 1777: 591–593[7]; Hoare [ed.] 1982: 473–474).

All these (and some other) motifs and episodes are well known from previously published works by the Forsters and they are also included in the *Mémoire*; there, however, they play a different role, serving more specified goals than simply enlivening the story to be told. In the *Mémoire* many of these motifs are meant to support the central message on the Rapanui society that the authors seek to convey to their audience. What should be stressed here, the way of treating the above-mentioned fragments of the analyzed document often bears an unmistakable stamp of George Forster's poetical and dramatic style, brought to exaggeration in the description of 'sad sea birds' and 'solitary whales' accompanying the *Resolution* in her struggle with bad weather and icebergs (*apud* Jakubowska 2014b: 75).

Due to the conciseness of the *Mémoire*, composed as an essay dedicated to a specific geographical location, chosen from among all the places that the Forsters visited during the Voyage, the manuscript resembles a condensed version of *Observations*. Although the fragments dealing with Rapa Nui are rather scarce in *Observations*, the island lends itself very well as an illustration of the philosophy and ideology that permeate the work by J. R. Forster. Two most prominent examples are: the climatic determinism developed by Forster (and present also e.g. in the philosophical thought of J. J. Rousseau[8]), and the theory of the Pacific peoples' origin, based on linguistic premises. The third example, this time pertaining to Easter Island in particular, is an attempt to explain the strikingly low number of women observed by the voyagers upon their landing and during the whole visit. Let us elaborate now on these questions.

According to J. R. Forster (and other philosophers/intellectuals, who are not considered in this paper), the climate and environment, in which a given tribe or society lives, influences people and their constitution, *cf. Observations*:

> Heat adds to the action of the heart a stimulus, and accelerates its pulsation; and since in a warm climate, the solids are more relaxed, than in a cold one, the impulse of the blood in the arteries finds less resistance, and therefore more powerfully expands the whole frame of the body [...]. This we find conformable to experience, for in hot climates mankind grow more powerfully, and attains earlier maturity and puberty. On the other hand, cold assuages the stimulus, and constricts the fibres, which naturally throws the whole system into a torpor or languid state. The heart does not act powerfully enough to carry on the functions with that vigour which is required, not only to accelerate growth, but likewise to overcome the greater resistance caused by the rigid state of all the solid parts. We find in consequence of these principles, the poor inhabitants of Tierra del Fuego, a small race of people, though descended from tribes, who, on the continent of South America, in a milder climate, and more happy circumstances, are very tall and athletic (Forster 1778: 177).

This situation holds true also for Easter Island:

> Heads of the islanders are very hairy, but their beards are not as thick as [the beards] of the inhabitants of Tahiti and neighboring islands, which may be attributed to the aridity of the soil and the dry climate of this island, since it is the humidity accompanied by heat that contributes to the beard growth, according to the observations of the best physicians (*apud* Jakubowska 2014b: 79–80).

[5]George Forster maintained that he quoted from his father's journal, but both versions are different.

[6]See above.

[7]See above.

[8]Cf. his *Discours sur l'origine et les fondements de l'inégalité parmi les hommes* (Discourse on the Origin and Basis of Inequality Among Men). 1755.

However, in consequence, all these circumstances influence also the culture originated in various regions of the world; here it is worth citing the *Mémoire*. Comparing the Rapanui art of *moai* monuments with Tahitian wooden counterparts (and considering the *moai* as much more developed), the author[9] of the document states:

> I recalled in my memory all the countries that cultivated fine arts with some success from their early days. The Upper Egypt, Attica, Italy – the cradle of arts, came to my mind as first. All those countries are dry and mountainous, but lie in a temperate and favorable climate. The Lower Egypt, located less favorably, developed the arts much later, and at present we can hardly find the remains of the most recent monuments, whereas those of the Thebaid, much more ancient [...] have survived [to our times] and still can be seen. Rich and fertile Boeotia, in a humid climate, surrounded by swamps and quagmires, was never famous for artists. Peatlands and swamps of Holland never gave birth to a great artist, and the efforts of the local genius never went beyond folk festivities, country weddings, landscapes and portraits. It seems that dry and raised climates [*sic*! – countries] of a temperate climate provide the human body with a more ardent and courageous constitution, which influences also their spirit and makes it prouder; on the contrary, low and marshy countries, of a humid and foggy climate, give birth only to soft and limp bodies whose influence on the mind diminishes proportionately the attachment to invention, the tendency to [figure out] great ideas and [the ability] of realizing them (*apud* Jakubowska 2014b: 89).

Contrary to their moderate, or even negative opinions expressed in the *Journal* and *A Voyage*, in the *Mémoire* the Forsters are praising the Easter Island art very highly. We are asking ourselves whether this circumstance – together with denigrating the Dutch art, for example – is something the naturalists really believe, or this attitude of theirs serves rather a different purpose. We are inclined to believe the latter is the case.

When it comes to the Pacific peoples' origin, the Forsters had no doubts as to their Polynesian roots. Not forgetting about such factors as external affinity among 'South Seas' islanders (skin colour, hair, etc.), or similarity of cultivated plants and manufactured artefacts, the naturalists put forward above all a strong and irrefutable argument: the language.

This is the domain of J. R. Forster; he is said to have known as many as 17 languages (Vorbrich 2011: 43), and clearly was a talented man in this respect. Both *Observations* and the *Mémoire* contain a comparative table of equivalence of the South Sea languages[10]; the naturalist writes:

> I took particular care in collecting the words of every peculiar nation we met with, that I might be enabled to form an idea of the whole, and how far all the languages are related to each other (Forster 1778: 185).

His disquisition on the affinity of languages and peoples throughout vast regions of the Pacific Ocean is impressive. Even more so when we think about a quite recent, well-known discussion on the alleged South American *versus* Polynesian descent of Easter Islanders, which was put to an end only by genetic research, and still not everybody is fully convinced. James Cook's crew had no such dilemmas, although – from our point of view – they did not possess precise tools and technologies to prove it. Let us quote just short fragments of the Forsters' lengthy and complicated argument, quite similar in both works, only slightly more concise in the *Mémoire*:

> ... we think it is an interesting aspect of the history of those islands, the most distant of which lie 1400 leagues apart. As we see that all along that enormous space the same language is spoken, we may therefore conclude that all the islands were occupied and peopled by the same human race. [...] We can see now that there are 1400 leagues from the Malacca peninsula to Islands of the Thieves, and from there, through the Carolines, to

[9]We should keep in mind that the Forsters presented the *Mémoire* as a work by Johann Reinhold solely.

[10]In case of the *Mémoire* 8 of them; in case of *Observations* 15, including some vocabulary from the native languages of the Americas.

> Island of Good Hope (or Hope Eylandt) [i.e. Niuafo'ou], the first of Friendly Islands, there are 1200 leagues; and from that island there are another 1400 leagues to Easter Island. And throughout all this area [that covers] 4000 leagues people speak dialects that are related to the Malayan language. [...] We would not be able to trace the migrations of these tribes without examining their language and comparing the harmony of the distant dialects. Only in this manner we managed to establish a new historical fact, that is at the same time a unique and striking phenomenon in the history of the human species (*apud* Jakubowska 2014b: 84–85).
>
> When a migrating race found in their new country, new birds, new fishes, and new plants, they must of course find new names for them, which could not exist in any other of the congeneric dialects. The qualities of these new animals, the preparation of the new vegetables into food and garments, required other new names, and thus it happened, that their language gradually assumed a distinct appearance, though after all, it was the same with that of the original tribe, being modified only to suit the organs of the tribe, and the new objects of their country (Forster 1778: 185).
>
> I am therefore rather inclined to suppose, that all these dialects preserve several words of a more ancient language, which was more universal, and was gradually divided into many languages, now remarkably different. The words therefore of the language of the South Sea isles, which are similar to others in the Malay tongue, prove clearly in my opinion, that the Eastern South Sea isles were originally peoples from the Indian, or Asiatic Northern isles; and that those lying more to the Westward, received their first inhabitants from the neighbourhood of New Guinea (Forster 1778: 190).

And now let us turn to the last question we mentioned above, namely the comparatively low number of Rapanui women. This puzzling situation was reported also during other early expeditions to the island. In *A Voyage* G. Forster attempts to give some explanation to this phenomenon. At first, he conjectures that the rest of women is kept in hiding in a remote region of the island or in caves, where the voyagers were not allowed to enter. But then he concludes that Cook's reconnaissance group has seen almost the whole of the island and the existence of a hiding place that could be omitted during that exploration is impossible. Although aware of rude manners of sailors, he also asks himself why should Easter Islanders have more reasons to be jealous of their women than Tahitians, visited earlier; and adds that Rapanui children – also few in number – have not been kept away.

'In short, this matter must remain unascertained' – George admits finally, giving up – 'and if, in fact, the number of women is inconsiderable, it must have been diminished by some extraordinary accident, which none but the natives could have explained; but, in all our doubts, our ignorance of their language prevented us from acquiring any information' (Forster 1777: 596).

Interestingly enough, in the *Journal* his father gives no single thought to this particular matter. However, he as well as George is of the opinion that in the past the island used to be more populated and happier. This conviction, present also in the *Mémoire*, underlies all posterior opinions according to which the *moai* must have been sculpted by some other tribe, more creative and civilized than the islanders met by the Europeans in the 18th century. This popular stereotype is one of the elements of the so-called collapse theory. All in all, J. R. Forster states: '... either a civil or external war, a great mortality, too great luxury, or some other disaster reduced those Islanders to the small Number we found them to be of' (Hoare [ed.] 1982: 476).

Meanwhile, quite unexpectedly, the same two authors give a neat, apparently logical explanation of the small number of Rapanui women both in *Observations* and in the *Mémoire*. From our modern and scientific point of view, however, their interpretation of the phenomenon is wholly Eurocentric and improbable:

> I suppose therefore, that as this isle has the strongest marks of having been once subject to a violent change from a subterraneous fire and earthquake, it is highly probably that in a great revolution of this kind, the numerous inhabitants of the isle were destroyed. Nor is this circumstance very improbable, for Capt. Davis in the year 1687, felt a violent shock of an earthquake in this ocean, and not very far from this island. In Otaheitee earthquakes are known and are thought to be under the regulation and conduct of *Maoowe*, a peculiar divinity: but this is rendered more probable from the practice of the inhabitants of Easter Island, who to this very day, frequently form their habitations underground, and support the whole fabric by walls of lose stones. Now if the disaster befell the nation in the day time, it is probable that a great many men being out of doors upon some business, might be preserved, whereas the women keeping more at home, were involved in instantaneous ruin, by the tumbling of the wretched habitations, and no more than a few only that happened to be out of doors were spared, to become the mothers or a future and unhappy generation (Forster 1778: 264–265).
>
> Since war cannot be a cause of the small number of women, as it is the men who expose themselves to danger, I supposed it was some kind of a universal and unexpected disaster that destroyed the majority of women of these islanders, what is all the more probable that the whole island was formed by a volcano; I knew that in 1687 the buccaneer Davis, on his way from Galapagos islands to Juan Fernandez, experienced a violent episode of earthquake in the open sea, which devastated Peru and totally ruined Callao, [and this had taken place] before he discovered the land that bears his name. Moreover, as I was told by the inhabitants of Tahiti that the South Sea islands were at a high risk of that calamity, I had no more doubts that an earthquake not only depopulated the island but also contributed to the decrease of the number of women. Since the islanders live underground and their women are busy with housework and [thus] attached to their homes, while the men work outside, it is very likely that one of those big, terrible natural disasters suddenly buried a great number of inhabitants, and, above all, a proportionally greater number of women (*apud* Jakubowska 2014b: 86).

It is a rhetorical question, to ask what the majority of Rapanui women could have been doing in the caves that could not have been done outdoors, in that hypothetical moment of a great earthquake. Anyway, the above-cited excerpts indicate another striking similarity between the French manuscript of the *Mémoire* and *Observations*.

The *Mémoire* and other Forsterian works: principal differences

It should not be a surprise for us that the *Mémoire* bears resemblance to other Forsterian works. What is far more baffling is the fact that the document comprises also certain discrepancies or even contradictions in comparison to – especially – *A Voyage* and the *Journal*. Let us name the most important ones.

It is a widespread opinion that no participant of the 18th century voyages that visited Easter Island saw *moai* quarries at Rano Raraku. G. Forster even states that explicitly in *A Voyage*: 'We neither met with any quarries, where they had recently dug the materials, nor with unfinished statues which we might have considered as the work of the present race' (Forster 1777: 594). N. Cauwe has even come up with a theory that the Rapanui, who willingly accompanied European visitors in their trips across the island, deliberately led them in such a way as to avoid showing the foreigners into local sacred places, such as the quarries (Cauwe 2011: 41–42).

Nonetheless, the *Mémoire* proves to be the only known document from the epoch of early European visits to Rapa Nui that states the opposite:

> If we consider the immense mass of stone, 27 feet high by eight [feet] wide, that probably was pulled out of quarries that we find only on the hills, in the middle of the island, where we saw during our [reconnaissance] trip some hats and some unfinished statues; [...] (*apud* Jakubowska 2014b: 87).

The mention of 'hats' (*pukao*) could indicate the Puna Pau quarry, which was alluded to by other reconnaissance trip members (e.g. W. Wales). However, the Forsters talk here also about more than one incomplete monument. Could that be a blurred memory or a mere slip of the tongue? It has to remain an open question.

A seemingly insignificant contradiction between the *Mémoire* and other works by the Forsters concerns the islanders' attitude towards entertainment. In *A Voyage*, commenting on the first encounter with a Rapanui who came on board the *Resolution*, George (Forster 1777: 600) concludes: 'We are unacquainted with the amusements of the people of

Easter Island, having never seen them engaged in any kind of diversion, nor taken notice of a single musical instrument among them.' In the manuscript, however, we find the following statement:

> If we are to believe that the [islanders'] behavior and outward manifestations should be correctly interpreted as [symptoms of] intentions incapable of hypocrisy, they all were happy and seemed to be deprived of any desire [...]. We saw them often laughing and playful, and several times even amusing themselves with singing and dancing (*apud* Jakubowska 2014b: 83).

This vision, contrasting with the previous quotation, but parallel with the Rousseauian concept of a 'good savage', fits exceptionally well with the image of Rapanui society that the naturalists are painting in the *Mémoire*. In spite of reporting cases of 'theft' or 'cheating' (misinterpretation of the Rapanui behaviour resulting from a different social code and value system developed in the Old World), the Forsters describe a happy, honest, morally sane and good-natured community. This community is living a simple and peaceful life under the rule of a wise and benevolent king, loved like a merciful father. Such a king, who is never condescending towards his subjects, deserves to be honoured and immortalized by monuments (i.e. the *moai*) erected out of free will of the islanders, and not out of fear or flattery. This paradisiacal interpretation of the Rapanui reality is contrasted with a vile, corrupted Europe, and complemented with a 'recipe' of a satisfactory everyday life, in the authors' opinion perfectly possible on Easter Island:

> ... if any young man wants to live on his own, he may simply occupy a [plot of] land and clear it out with a tool made of hard wood, in form of a pointed stake, which is used here instead of spade; his family and friends will not refuse him some sweet potatoes that he will divide into cuttings to multiply them; others will give him some shoots of sugar cane or banana tree, and there he is, ready to become head of a family and to feed himself, his wife and children. A hole dug in the ground and strengthened by stones serves as a shelter from the heat and at night. A couple of stems of cultivated mulberry supply him with the bark out of which his wife makes the scanty clothing he needs. A [sort of] grass supplies him with fiber to make strings or cords. He uses shells [...] to make hooks that serve him to catch fish. His farm consists of a pair of chickens and, as he is not picky, he also tries to catch frigates, gannets, brown noddies and other seabirds that nest on the rocks off the shore, and also does not deny himself even rats; in this way he gets rid of them as enemies of his plantations and at the same time treats them as a delicacy. Water from several wells existing on the island is his usual drink [...] (*apud* Jakubowska 2014b: 83).

In a way, the Forsters bring to life the primeval, Arcadian vision of Easter Island. The vision that is also evoked by modern authors, only to be shattered by a phantom of the so-called 'collapse' or 'ecocide' theory, much disputed in recent years. But this apparent paradise contrasts also with the disappointment of Cook's crew and the Forsters themselves, present in their other works. In our opinion, the reason is a political one.

The political rationale behind the *Mémoire*

It must be remembered that the only Forsterian text created during Cook's Second Voyage is the *Journal* written by Johann. As we have already mentioned, Johann Forster believed he had the oral promise to be the main, if not the sole author of the narrative on the expedition. Nevertheless, after the voyage had ended, things turned out differently. Literary ambitions of James Cook, as well as political and ideological differences between the anti-imperial J. R. Forster and the captain, can be cited among the reasons that led to a serious controversy: 'some of the most influential Tories in the British Admiralty tried – aggressively, unceasingly, and successfully – to thwart the literary and scientific output of the Forsters' (Vorbrich 2012: 90). Johann Forster was prohibited from publishing his narrative, but this ban did not concern his son. That is why George issued his excellent narrative *A Voyage*, which – as we already know – sparked further debate[11].

Disenchanted by the atmosphere in Great Britain, coping with financial problems, the Forsters decided to leave for the continent. As mentioned above, George travelled to Prussia around the year 1779, to secure for his father a professorship in Halle (Vorbrich 2011: 62). As Marzena Chrobak points out (2010: 11) Frederick the Great, the monarch highly

[11]A detailed description of the situation can be found in Vorbrich 2011, 2012.

praised in the final fragments of the Mémoire, helped them and also helped to bail out Johann Forster from prison. This French-speaking king, believing himself to be a servant of his nation, and not a God's anointed, a sovereign known as a 'philosopher from Sanssouci', is the most probable addressee of the finely prepared manuscript. In these circumstances it is no more surprising that the *Mémoire* is signed by Johann Reinhold Forster alone.

Geographical and navigational problems exposed in the *Mémoire* and other relevant works

Apart from literary and historical reflections, we would also like to address an issue rarely elaborated on in the context of Rapa Nui. Namely, we feel it is time to examine closely the geographical position of Easter Island (as well imaginary, as a real one), and the navigational problems it posed in the times of early maritime visits. In the *Mémoire* written by the Forsters there are some excerpts that seek to explain the question of the then unreliable coordinates of the island. '[W]e discovered it on the eleventh of March of 1774, and we found it located at 27° 5' south latitude and at 109° 46' of longitude west of Greenwich' (*apud* Jakubowska 2014b: 74), the authors write somewhat boastfully, in the name of James Cook, apparently. Moreover, explaining briefly the history of 18th century expeditions, and doubts reigning among the voyagers, they add two footnotes:

> Easter Island lies more than 707 leagues from Copiapo, and, consequently, Davis Island (that lies only 500 leagues from there) is situated at least at eleven degrees more to the east than Easter Island; thus it is evident that Roggewein was first to see it, and I was convinced by indisputable signs that the Spanish were here before us, although they believed that they had touched upon Davis Island. The sandy islet of Davis certainly lies more to the east, and Mr. Bougainville left it to the south of his route, as Captain Carteret left it to the north of his.
>
> [...]
>
> From l'histoire des Gouverneurs de Batavia we know that Roggewein situated Easter Island at 265° 42' from Tenerife = 110° 45' west of London, so the difference is only of 59'. The Spanish placed it at 268° 19' from Tenerife = 108° 11' west of London. The Dutch affirm the latitude of the island at 27° 4' S and the Spanish at 27° 16' of the south latitude (apud Jakubowska 2014b: 74)[12].

When Cook's Second Voyage started, the geographical situation of Easter Island was far from clear. Even the 'identity' of the island was unsure. There was still uncertainty concerning the alleged, or even imaginary Davis Land: Does it exist or not? Is it the same as the 'Paasch Eyland' discovered by Roggeveen in 1722? Or are there two islands? Maybe one of them is not an island, but the much searched for Southern Continent? Many conjectures, but few truths.

We have to remember that Easter Island was not an important port of call for James Cook. It was just a 'tiny speck of land' – as it is often referred to – difficult to anchor at, that, at a certain point of the voyage, proved to be worth visiting in search of fresh food or water. Or, that was what some of the crew thought, having read only unreliable reports from the Dutch voyage. *Summa summarum*, it was not a goal in itself – but some navigational aspects related to the British visit on Easter Island are crucial. Let us first explain the situation.

It is a well-known fact that the Dutch expedition that discovered Easter Island for the Old World turned into a fiasco[13]. Roggeveen did not find the Southern Continent, lost one of is ships and was arrested due to entering the area under the Vereenigde Geoctroyeerde Oostindische Compagnie (VOC)[14] monopoly (his fleet was equipped by the Geoctroyeerde West-Indische Compagnie or WIC)[15]. As a result, all official documents of the voyage were confiscated and subsequently lost[16]. Therefore, Cook could not obtain any credible information about Roggeveen's route, findings, or discoveries. The only texts the British expedition members could read were the memoirs by Behrens (German edition:

[12]The authors give the Spanish coordinates wrongly; cf. our Table 2.

[13]The whole story is told by Van Gelder (2012).

[14]Dutch East India Company.

[15]Dutch West India Company.

[16]The Roggeveen's journal was found and published in 1838; the journal of one of his captains, Bouman, appeared in 1910 and was published in 1911. The fate of two other journals, written by captains Koster and Rosendaal, is unknown.

1737; English edition, translated from French, in Dalrymple 1771), of a very dubious value, or 'anonymous' reports being actually compilations of simple sailors' fantasies (Dutch editions: 1727, 1728; English edition in Dalrymple 1771)[17]. There were also Spanish documents produced during the voyage of Felipe González in 1770[18]; however, Cook himself admits he does not know them, but he 'was told' that the Spaniards visited Easter Island in 1769 (*sic*) (*apud* Beaglehole [ed.] 1961: 342). The Spanish were not too keen on revealing any information concerning their activity in the Pacific Ocean, as they preferred to keep it secret (Mellén Blanco 1986: 21–22; El Diario Montañés 2007). The voyage to Easter Island in 1770, ordered by the viceroy of Peru, Manuel Amat y Junyent, was of military nature. González obtained detailed instructions concerning his obligations (*apud* Mellén Blanco 1986: 213–219); it is worth mentioning that the Spanish did not take into consideration the previous Dutch presence on the island or the toponym applied to it by Roggeveen; they still used to speak of 'las tierras, o Ysla conocida con el nombre de *David*' ("lands or island known under the name of David" – *sic*). Strikingly enough, in the Spanish reports the only allusion to the Dutch maritime activity in itself appears in form of two mentions of a certain Van Keulen map[19]. The above-mentioned instructions warned González:

> *Lo que mas podria influir al malogro de la expedición [...], es, la discrepancia, que se nota de la longitud en que colocan situada esta tierra de David mas de cincuenta Cartas de diferentes Naciones, que se han tenido á la vista para formar esta instrucción* [...] (*apud* Mellén Blanco 1986: 215)[20].

It seems that, after the alleged 'discovery' of what was called Davis Island in 1687 reported by Wafer (*apud* Dalrymple 1771: 123), many cartographers – guided by a sort of wishful thinking – included the island on their maps, as a possible sign of a nearby Southern Continent. Only the latitude of the island was more or less accurate; as the instruction for González puts it, between 27° and 28° South (Wafer gives 27° 20') – but the longitude was a controversial matter. Wafer says only that the island 'bears from Copayapo [= Copiapo] almost due E five hundred leagues'. It seems that as well the Dutch under Roggeveen, as the Spanish under González, used dead reckoning to establish their longitudinal position; Francisco Aguera of the Spanish crew gives the most detailed information about his methods of measurement (*apud* Mellén Blanco 1986: 295–336). All in all, Cook had no reliable geographical data for Easter Island and admitted he harboured no real hopes to reach it. In two versions of the same journal record from 6th February 1774 he writes: '... Easter Island, the situation of which is so variously laid down that I have little hopes of finding [it]' (*apud* Beaglehole [ed.] 1961: 326). And: '... then to go in Search of Easter Island or Davis's Land, whose Situation is known with so little certainty that the attempts lately made to find it have miscarried' (*apud* Beaglehole [ed.] 1961: 327).

Let us now analyze more thoroughly Cook's journals from the penultimate year of his Second Voyage – 1774 – and namely the time span of one month and eleven days (41 days in total) from the day the *Resolution* reached her most Southern point of latitude 71° 10' on Sunday 30th January, until Easter Island was seen from the top of the mast on Friday 11th March. Our aim is to examine – in the background of the Forsters' French manuscript – both the chain of events, which led Cook to the re-discovery of Easter Island, and the navigation methods used by him in the process. We close by insisting that, bearing in mind the

[17]They also quote other documents, e.g. a letter by Dalrymple.

[18]Of them copies, excerpts, and some originals have survived; these documents, almost in their totality, have been transcribed by Mellén Blanco (1986, 1988).

[19]'El Día 21 de Noviembre de 1770 Miercoles [...] me puse a la vela [...] a fin de ponerme en 260 grados 20 minutos que es la Longitud en que la carta holandesa pone otra Ysla con el mismo nombre de David, para quedar cerciorado de si son dos ó solo una' (González *apud* Mellén Blanco 1986: 225). – 'El Miercoles 21 [...] nos hicimos a la vela [...] costeando la parte del N. de la Ysla en demanda de otra que pinta la carta Olandesa de Wankeullem [sic]' (Aguera *apud* Mellén Blanco 1986: 312). Translation: González – 'On the 21st of November, Wednesday [...] I set sail [...] in order to reach 260° 20', which is the longitude under which, on the Dutch map, there is this other island, named David; I wanted to determine if there are two of them or only one'. – Aguera – 'On Wednesday the 21st [...] we set sail [...] following the North coast of the island, in search of another one, visible on the Dutch map by Van Keulen'. All quotes translated by ZJV.

[20]'What could contribute to the failure of this expedition [...] is the discrepancy in the longitude under which the land of David is located on more than fifty maps, published by various nations, that were examined in order to issue this present instruction [...]'.

scarcity of the positional information Cook got on the piece of land he reached, his seeing the land has every vestige of re-discovery or even discovery in the proper sense of the word, backed by the 'sixth sense' and unprecedented navigational skill of the commander.

The principles of the determination of the latitude by means of the sextant or other similar angle measuring instrument, the determination of the longitude by means of the dead reckoning (further on: D.R.) and a clock (chronometer), as well as the methods used, were described in Vorbrich 2011 and they will not be repeated in the present article[21].

However, we would like to point out the main drawback of the clock method: it requires minimum three clocks. With one clock there is no notification if its rate has changed. If there are two clocks, the navigator has no way of knowing which one is (or both are) correct or wrong. If you have at least three clocks, you may eliminate one of them as being in error (if it is in error) using the so called 'majority voting' method. However, during Cook's Second Voyage it was deemed necessary to repeatedly check the clock against a non-clock method of longitude determination, such as Galileo-Cassini Jovian and/or lunar distance measurement.

What remains here is to shortly describe the Galileo-Cassini Jovian and the lunar methods of longitude determination. We will be mostly interested in the history of the potential navigational application of these methods, spanning the time between the discovery of Easter Island in 1722 by Jacob Roggeveen and its re-discovery – in a navigational sense – in 1774 by James Cook.

The first mentioned longitude determination method employs Jupiter's moons. We consider that the best explanations were provided by the original author of the method. In 1612, soon after Galileo discovered Jovian satellites, he proposed the method of determining terrestrial longitude by means of timing of the eclipses of the moons by Jupiter. This idea attained practical accuracy when in 1668 Giovanni Domenico Cassini published the first ever, state-of-the-art, specialized astronomical tables of the Jovian satellite Io, titled: *Ephemerides Bononienses Mediceorum syderum et hypothyesibus et tabulis Io* (Westfall and Sheehan 2015: 191). Io, whose orbit is the closest to Jupiter, is particularly handy for the purpose of timing[22]. For the present paper readers who happen to be practical enthusiast astronomers and own even the simplest telescope, it is obvious that in order to apply the Galileo-Cassini method the deck of the ship has to be almost stationary during the observation[23]. It happens very seldom. However, the Cassini's tables have been used with success on land. For example, already in 1669–1670 Io simultaneous observations, in Paris and in Tycho de Brahe's astronomical observatory in Uraniborg, led to the longitude measurement, which agrees with the present-day data within 0.4°, equivalent to 16 nautical miles[24].

Theoretically, time-wise, Cassini's *Ephemerides [...] tabulis Io* might have been available to Jacob Roggeveen. However, he paid a single-day visit to Easter Island, he did not stay there during the night, therefore, intrinsically, he was not in a position to determine its longitude by means of the Galileo-Cassini method. The lunar distance (for short, 'lunar') method of longitude determination involves the determination of both the Greenwich and local time. The determination of the Greenwich time

[21]Further on we write latitude/longitude in the narrative text and LAT/LON respectively when used in equations or tables.

[22]Io's orbital period is approximately 42.5 hours.

[23]The observer has to look at Jupiter through a powerful telescope. If the ship's deck moves even a tiny bit, Jupiter may disappear from the observer's vision. In addition, it is advisable to collect data from at least two eclipses of Io, which in practice requires that ship does not resume her voyage for the timespan of the orbital period.

[24]Note 1. Uraniborg (on Hven, an island in the Øresund) coordinates (GPS): 55.9079° N, 12.6961° E.
Note 2. The value of 0.4° is cited after Westfall and Sheehan (2015: 192).
Note 3. 1 nautical mile = 1852 metres.
Note 4. To scale the units of arc to units of time, please use the following indications:
360° in Earth rotation is equal to 24 hours of time; 15° in Earth rotation is equal to one hour of time; 1° in Earth rotation is equal to four minutes of time; 0.5° in Earth rotation is equal to two minutes of time. Therefore, in 1670 the state-of-art Io occultation observations were conducted with the error of time of app. 2 minutes. At the equator LON error of 1° = 60' is equivalent to 60 nautical miles.
At the LAT of Paris equal to app. 49°, LON error of 1° is equivalent to cos (49°) x 1852 m x 60° is equal to 0.66 x 1852 metres x 60' = 73.339 metres.
Summa summarum: using Cassini's tables, the Io occultation observations for a LAT of app. 50° are expected to generate an error in LON of ca. 30 kilometres, equivalent to 16 nautical miles.

requires lunar tables, which are listing the position of the Moon in the celestial sphere *versus* time, and an angle measurement tool such as sextant[25]. The measured angles are: between the Moon and a celestial object (called 'lunar'), and between the horizon and these two objects. The more accurate the Moon tables are, the more accurate are the time determination results.

Local time can be determined from a sextant measurement of the altitude of the celestial object. The history of the idea of longitude determination by means of the lunar distance (first mentioned in 1514 by Johann Werner) is time-wise connected with the history of the idea of longitude determination by means of a clock (first mentioned in 1530 by Gemma Frisius).

Both 'lunar' and clock methods of longitude determination had a common factor, which increased the pace of the research on the improving of the accuracy of the lunar tables. It was a prize proposed in 1714 by the British Parliament (so-called 'Longitude Act') granting a huge sum of money for an author of a method enabling determination of longitude at sea within the accuracy of half arc degree, and half this sum for the accuracy twice worse (Porter [ed.] 2003: 341).

For a long time both methods remained impracticable from the point of view of the oceanic navigation. By middle of the 18th century the Moon tables computed by theories of three of the greatest minds of the epoch: Clairau, Eurer, and d'Alambert, dated 1747, 1751, and 1754 respectively, could not meet the above-mentioned standard, and at sea gave an error of the longitude determined by means of 'lunars' bigger than 1 arc degree (Porter [ed.] 2003: 339–341). In 1753 Tobias Mayer (a German astronomer) published lunar tables that gave an error that met the lower accuracy requirement of the 'Longitude Act' of less than one arc degree.

It so happened that in 1765 the British Parliament awarded the highest prize to John Harrison, for a clock, which gave an error lesser than half arc degree.

As already said, Cook – during his Second Voyage – had on board tools for the state-of-art longitude determination by means of both methods. However, back in in 1722, Roggeveen had none of them. We have to assume that for Roggeveen the main tool for longitude determination was D.R., which might give unpredictably useless results[26].

From one entry of Cook's Journals (on 19th February), we learn that, for the longitude determination, apart from the clock method, he used the then state-of-the-art method in lunar astronavigation, with its variation, which involved the simultaneous observation of the Moon and the Sun (a relatively rare phenomenon)[27]. For each of the 41 days in question Cook provides information about his latitudinal and longitudinal position. Although he does not always mention it explicitly, we assume that he always refers to the local astronomical noon (further: l.a.n.) position. Cook explicitly mentions stand-alone[28] D.R. as a tool for longitude determination at l.a.n. eight times, that is, on the last two days of January and the consecutive first six days of February. Once (on 8th March) the

[25]For the history of the lunar tables and associated accuracy of LAT determined by means of them, cf. Porter [ed.] 2003.

[26]We cite fragments of an email by Roelof van Gelder, the biographer of Roggeveen, received in June 2016: 'In Roggeveen's time they hadn't good, working, reliable clocks so they had to cope with hour glasses. [...] I think Roggeveen worked with printed moontables, but the first *exact* moontables were only published after Roggeveen's time by the German Johann Tobias Mayer. [...] The real problems started in the Pacific. But there they kept (starting from Juan Fernandez) a more or less stable course westwards, south of the equator (ca. 27 degrees south). The basic idea (and their official task) was: if they just kept this latitude (which they could keep rather easily), they would finally and automatically find (after 500 miles) what they were looking for: *Terra Australis Incognita*. After these 500 miles they had to sail on a 20 degrees course south of the equator westwards. Besides that: it is good to remember that Roggeveen's father and brother were both cartographers and Jacob Roggeveen himself must have had a kind of training in navigation and cartography. And moreover: each of the three ships had experienced pilots on board'.

[27]Note 1. 'SATURDAY 19*th* [...] At 3 o'Clock in the PM our Longitude p^r^ Mean of Several Lunar Observation [Footnote by Beaglehole: "by Mr Wales, Clerke, Gilbert and Smith"] was 94'19'30" [*sic*, spelling error – should be 94°] West, the watch at the same time gave 94° 46' w [*sic* – W]' (*apud* Beaglehole [ed.] 1961: 330). Those observations gave the difference of LON ('lunars') minus LON (clock) equal to app. 26', or app. ½° of LON.
Note 2. The total error of the lunar distance method, given the state-of-art lunar tables and very experienced observers, is supposed to be in the region of ¼° of LON, equivalent to 15 nautical miles at the equator.

[28]'Stand-alone' is a present-day term used in navigation.

captain, referring to longitude determination at l.a.n., mentions D.R. compared with a clock[29]. For the remaining 32 days he does not mention any method he used for l.a.n. longitude determination and we presume that the stand-alone clock was used as a primary tool[30]. However, we allow that in the above-mentioned period of 32 days Cook's astronomer and officers might have made and recorded observations of 'lunars', and Cook (or his editor Beaglehole) might not have mentioned them explicitly in his journal – published in 1961 (which is not a primary source *sensu stricto*)[31].

The clock as a tool for longitude determination was totally new for the British Admiralty and therefore for Cook. The captain could not consider the clock hundred percent reliable and having a stable rate. The perusal of Cook's journals for the period outside the above cited 41 days points out that, whenever practicable, Cook – while sailing – compared different longitude determination methods such as clock, D.R., and 'lunars' methods[32].

During the above-mentioned 41 days in his navigation decision making process Cook was distracted between two seemingly disparate aims, namely making discoveries for the Crown and the speediest possible finding a piece of land to give respite to his cold-weary crew. Although Cook wrote that his crew was healthy and welcomed with glee the perspective of prolonging the voyage for another year (*apud* Beaglehole [ed.] 1961: 327–328)[33], the Forsters wrote something diametrically different: about the crew almost dying from cold, suffering from scurvy[34], as if accusing Cook of callousness. In their eyes the captain showed his true self by continuing sailing in cold waters seeking his own fame instead of speedily returning to a warmer climate[35].

Cook and his crew hoped that Easter Island, a.k.a. Davis Land, could procure food and water. However, Juan Fernández land (still another name for the Southern Continent; not to be confounded with Juan Fernández archipelago), if found, could possibly procure both supplies and more fame. So Cook went first in quest of Juan Fernández. Only when he made sure that he could not find it, did he sail for Easter Island, a.k.a. Davis Land.

The problem of Easter Island *versus* Davis Land again requires a moment of thought. The longer we study Cook's journals, the more we are inclined to think that for him, and for the professional discoverers like him, apparently the name of Davis Land weighed more than the actual geographical position of it. It cannot be rechecked whether Davis ever saw his alleged land, different sailors assigned it different coordinates, sometimes hundreds of miles apart. The name of Davis Land survived as a promise of the long-sought-for Southern Continent.

To our mind, looking for Davis Land was a sort of senseless mania. Several tiny specks of land, apparently seen in the Pacific, were identified with

[29]'TUESDAY 8*th*. [...] [apparently noon] [Lat.] 27° 4'. [Long. by reckoning] 103° 58'. [Long. by watch] 105° 3' ' (*apud* Beaglehole [ed.] 1961: 336). The longitude determined by means of D.R. – LON (D.R.) – was apparently counted during 24 hours, i.e. from the last noon longitude determined by means of the clock – LON (clock). Those observations gave the difference LON (D.R.) minus LON (clock) equal to approximately 1° 5' of LON.

[30]Navigation with a single tool such as a clock is currently called 'sole means navigation'.

[31]Note 1. Beaglehole offers an additional editorial explanation: 'In H Cook has written further observations for the dates 22 February – 11 March. [...] The additional figures note the longitude by watch [...]' (Beaglehole [ed.] 1961: 331).
Note 2. H: MS Journal (transcript) in the possession of Lieut.-Commander Palliser A. Hudson (Beaglehole [ed.] 1961: xv).

[32]As pointed out in Vorbrich (2011), whenever the *Resolution* stopped near land for sufficient time, Cook's astronomer Wales established an observatory on land and used the Gallileo-Cassini method not only to improve the accuracy of the land-based LAT determination, but also to check the clock's drift. Cook made the final check of the clock in 1775, nearing Plymouth and comparing the LAT (by clock) and the true LAT of his departure point in 1772.

[33]'SUNDAY 6*th* [...] a healthy crew and not in want of either Stores or Provisions [...]' (*apud* Beaglehole [ed.] 1961: 326–327). – '[...] the Seamen [...] rejoiced at the Prospect of its [Voyage] being prolonged a nother year [*sic*]' (*apud* Beaglehole [ed.] 1961: 328).

[34]'[1774] Jan, *y^e^* 20*th* [...] bread [...] entirely rotten [...] not enough for the support of a [...] man [...]. The Doctor expects that if our this years cruize [*sic*] will prove so long as that of last year, to see half of the Ship's company sick' (*apud* Hoare [ed.] 1982: 448).

[35]'[1774] Jan, *y^e^* 5*th* [...] there are people [= Cook], who are hardened to all feelings [...] by their perseverance, which costs the lives of the poor Sailors or at least their healths' (apud Hoare [ed.] 1982: 443–444).

it. As we already know, the same happened with the true Easter Island. Cook might have chosen the coordinates to sail to, from among disparate data given on various maps, but he had been restrained by wind direction, so it is sometimes difficult to guess the correct position he aimed at. His decisions shown against the time scale of 41 days are as follows:

On 30th January Cook takes a tactical decision not to continue the Voyage South[36].

On 6th February Cook takes a strategic decision to prolong the Voyage into the next year (1775) and makes tactical plans concerning Juan Fernandez and Easter Island, a.k.a. Davis Land[37]. He proclaims his resolution to spend the Southern Winter 1774 (Northern Spring) in the tropics.

On 22nd February Cook pronounces that 'it is probable that the whole [discovery of Juan Fernández] is a fiction and that no such discoveries were ever made.' He gave up his search for Juan Fernández and 'stood away to the North in order to get into the Latitude of Easter Island' [Cook does not write down this target latitude, he does not write the target longitude either].

On 26th February Cook writes: 'At Noon steer'd NNW in order to get into the Long^de^ of Easter Island'.

On 5th March Cook states: 'the many birds we see [...] we are in hopes of meeting with Davis or Easter Island'.

On 9th March Cook makes an important approach maneuvre and writes: 'Judgeing our selves by observation to be nearly in the Latitude of Davis's Land or Easter Land we steer'd nearly due west'.

On 11th March they saw land; Cook wrote: 'made sail and soon saw the Land from the Mast head bearing West' (*apud* Beaglehole [ed.] 1961: 322–336).

Johann Forster's entries in his *Journal* in the above mentioned 41 days period encompassing the approach to Easter Island are scanty and we are inclined to believe that he was not admitted to the secrets of Cook's decision-making process. Cook revealed his new strategy (to prolong the Voyage for another year) to his officers with some delay. Until then the officers and crew were not aware that Cook contemplated not to sail to Cape Horn in 1774, so for them the sailing to Easter Island was not even in the cards. The Forsters may have learned the details even later. Prior to that Johann wrote: '[1774] Jan ye 5th [...] it would be best to make best of this fine gale, & bear down again into 60° & upwards to round Cape Horn' (*apud* Hoare [ed.]1982: 444).

[36]The *Resolution* sailing South reaches the furthest South LAT of his expedition: 71° 10' and gets to a place where: 'The outer or Nothern [*sic*] edge of this immence [*sic*] Ice field was compose[d] of loose or broken ice so close packed together that nothing could enter it [...]'. Cook decides to turn back North, reasoning: 'I will not say it was impossible anywhere to get in among this Ice, but I will assert that the bare attempting of it would be a very dangerous enterprise' (*apud* Beaglehole [ed.] 1961: 321–322).

[37]'steered [...] with a resolution to proceed directly to the North, as there was no probability of finding Land in these high Latitudes [...]' (*apud* Beagehole 1961: 325). – '[...] my intention was, first to go in Search of the land, said to have been discovered by Juan Fernandes [*sic*] above a century ago [footnote by Beaglehole: "This is not the actual island Juan Fernandez discovered [...] in 1563 [...] but the continent he was alleged to have found some years later [...]".] in about the Latitude of 38°; if I failed of finding this land, then to go in Search of Easter Island or Davis's land, whose Situation is known with so little certainty that the attempts lately made to find it have miscarried [Beaglehole cites Cook's other remarks: "positions laid down by Dalrymple and others (lat. 27° s, long. 106° 30' w) and Pingré (lat. 28° 30' s, long. 123° w)]' (*apud* Beaglehole [ed.] 1961: 326–327). To our mind, 18th century mariners sailing towards land having the coordinates of the present-day Easter Island had to know the approximate coordinates of an important reference point, Copiapo, on mainland Chile (present-day coordinates: 27° 22' S, 70° 20' W). Anyway, Cook must have known them.
The present-day coordinates of Easter Island are 27° 06' S, LON 109° 21' W. Cook's measurements in 1774 were as follows: 27° 5' S, 109° 46' W of Greenwich.
The difference LON (Copiapo) minus LON (Easter Island) is app. 39°.
The difference LON (Darlymple) minus LON (Pingré) is app. 17°.
The value of 17° is equal to app. 43% of 39°.
We consider that Cook must have known that the D.R. error of app. 40% accumulating at a distance of app. 40° in LON had been rather highly improbable both in 1687 (Davis) or in 1722 (Roggeveen). Therefore, Cook must have known that LON (Dalrymple) and LON (Pingré) cannot concern the same piece of land. However, in 1774 Cook was not sure whether at the LAT belt from app. 27° to 28° 30' S there was no land relatively close to the West coast of South America in the longitudinal belt app. 17° wide.

No.	Date	LAT†	LON‡	Wind	Course	Remarks
01	25	37° 50'	101° 08'	North to NW	S 78° 15' W	NO
02	26	36° 37'	101° 57'	SW	NNW1/2W	NO; steered to get into LON of E. I.
03	27	34° 53'	102° 33'	SW to SSE	N 15° 30' W	NO; 'now taken ill [...] to my bed' *
04	28	33° 7'	102° 33'	SE to NE	North	Wind SE NO, wind NE YES
05	01	32° 8'	102° 47'	Easterly to SW	N 7° W	NO
06	02	31° 12'	102° 29'	WSW to WNW	N 15° E	Wind WSW NO, wind WNW YES
07	03	30° 37'	101° 45'	NW & NNW	N 47° E	Wind NW NO, wind NNW YES
08	04	29° 56'	100° 59'	NNW, WNW	N 46° E	Wind NNW YES, wind WNW NO
09	05	29° 44'	100° 45'	NW & calm	NE	NO; hopes meeting Davis or E. I.
10	06	29° 23'	101° 3'	NE	N 38° W	NO
11	07	28° 20'	102° 3'	NEbE, East	N 40° 15' W	NO
12	08	27° 4'	103° 58' R 105° 3' C	Easterly	N 52° W	NO
13	09	27° 7'	106° 00'	Easterly	W 2° S	FOL; nearly LAT Davis or E. I.
14	10	27° 9'	107° 55'	Easterly	West, Southly	FOL
15	11	27° 11'	109° 2'	Easterly	-	FOL; 'saw the land' **
16	-	27° 5'	109° 46'	-	-	E. I. (*apud* Jakubowska 2014b: 74) ***

Table 1. Cook's approach to Easter Island (E. I.), February – March, 1774, according to his journal.

Legend

According to our comparison of wind and course (from Cook's journal):
sailing beating hard against the wind: NO or YES,
sailing with the following wind: FOL;
According to Cook's journal:
LON R = longitude determined by means of dead reckoning (D.R.),
LON C = longitude determined by means of clock.

† South
‡ West of Greenwich
* 'I was now taken ill of the Billious colick and so Violent as to confine me to bed, so that the Management of the Ship was left to Mr Cooper my first Officer' (apud Beaglehole [ed] 1961: 333).
** '[…] saw the Land from the Mast head bearing West […] At Noon […] Distant about 12 Leagues' (apud Beaglehole [ed.] 1961: 336).
*** LON 106° 00' (probably predetermined by Cook to be reached) minus LON 109° 46' = 3° 46'.

Year/name	LAT	LON from Tenerife	LON from Greenwich	Remarks
1687, Wafer	27° 20'	-	-	500 leagues E of Copiapo
1722, Bouman	27° 27° 6'	258° 52' 260° 8'	-	calculated & acc. to Van Keulen map calculated by Bouman
1722, Behrens	28°	263°	-	unreliable, a plain soldier on board
1770, Hervé	27° 06'	264° 20'	-	one of the tips
1770, Hervé, maps	27° 6'	264° 30'	-	a note on one of the maps (different ink, same hand): Longd. O de Cadiz 105° 55' El Capt. Cook la vio en 1774 y dio de [?] Longd. de Cadiz 103..30 Diferencia 2° 25'
1722, Roggeveen	27° 4'	265° 42'	-	George Forster 110° 45' from London
1770, Aguera	27° 2'	267° 1'	-	E tip
1770, González	27° 2'	268° 30'	-	Instructions: reach 264°
present coordinates	**27° 11'**	**-**	**109° 35'**	
Cook's target	27° 7'	-	106° 00'	Coord. 9th March 1774, noon, 3 days sailing to E. I. Wales: 27° 07' 3 miles from SE tip; 250° 08' 17½" E of Greenwich (*sic*)

Table 2. List of LON of the lands called Davis/David's Land and/or Easter Land, which Cook might have considered as a target.

Below we cite the relevant entries of Johann Forster's *Journals:*

> [January 1774]
>
> 30th [...] I believe, it is so far South, as ever any man in future times shall choose to go; it being nearly 71° South.
>
> [February 1774]
>
> 26th [...] We sailed on in the Course NWbW. [...] The Latitude at noon 36° 48' [arc min] South.
>
> 27th [...] We altered the course at noon & stood on NWbN.
>
> 28th [...] We now go directly for *Easter-Island*, & every one in the Ship most ardently wishes to see this Island, in hopes of getting a good many refreshments of which we stand much in need, having been more than 13 weeks out at sea.
>
> [March 1774]
>
> 9th [...] Altered the Course to W½S, a fine fresh breeze setting us forwards towards the longexpected [sic] & wished for *Easter Island*.
>
> 10th [...] Our expectation increased to see *Easter Island*.
>
> 11th [...] The weather cloudy: we are all in hopes to see the Land, which would be no doubt an agreable [*sic*] sight; after being out at Sea during fifteen weeks. [...] During part of the night we brought to, at 5 o'clock, when we began to make sail we saw the land from the Mast head, it lay a point to the North of our Course or something less: it seems to be about 15 leagues distant: two hummocks are seen & to the North the land goes down & runs out into a [*sic*] low point (*apud* Hoare [ed.] 1982: 451–460).

George Forster's description of the above-mentioned 41 days period cannot be considered as a primary source *sensu stricto*, as it was written after the *Resolution* had returned. In our opinion, George – now having a possibility of checking sources of information unavailable to him while on board the *Resolution* – even committed the so-called error of presentism, whether deliberately or not. Below we are presenting George's view on this subject, published in *A Voyage*:

> We passed the night in making several trips, in order to keep to windward of the island and as near it as possible, resolving to pursue our search of anchorage the next day. In the mean time we reflected on the excellent means of ascertaining the longitude, with which our ship had been furnished, and which had carried us exactly to this island, though several former navigators, such as Byron, Carteret, and Bougainville had missed it, after taking their departure from islands at so short a distance from it as those of Juan Fernandez[38]. Captain Carteret it seems was only misled by an erroneous latitude in the geographical tables which he consulted; but this could not be the case with the rest. We had the greatest reason to admire the ingenious construction of the two watches which we had on board, one executed by Mr. Kendal, exactly after the model of that made by Mr. Harrison, and the other by Mr. Arnold on his own plan, both which went with great regularity. The last was unfortunately stopped immediately after our departure from New Zeeland in June 1773, but the other went till our return to England, and gave general satisfaction. It appears, however, that in a long run the observations of distances of the moon from the sun or stars, are more to be depended upon, if they be made with good instruments, than the watches or time-keepers, which frequently change their rates of going. The method of deducing the longitude from the distances of the sun and moon, or moon and stars, one of the most valuable acquisitions to the art of navigation, must immortalize its first inventors. Tobias Mayer, a German, and professor at Gottingen, was the first who undertook the laborious task of calculating tables for this purpose, for which his heirs received a parliamentary reward. Since his death the method was so much facilitated by additional calculations, that the longitude will perhaps never be determined with greater precision at sea by any other means. The latitude of Easter Island corresponds within a minute or two with that which is marked in admiral Roggewein's own MS. journal[39], and his longitude is only one degree erroneous, our observations having ascertained it in 109°46' west from Greenwich. The Spanish accounts of the latitude are likewise exact, but they err in longitude about thirty leagues (Forster 1777: 554–555).

To sum up: our analysis of Cook's journal and of Fosters' writings leave us in the dark as to why Cook decided to intentionally reach the longitude – which he considered to be that of Easter Island – at a certain latitude positioned two days of sailing from the true longitude of Easter Island. We presume that the captain could not have known the true longitude of Rapa Nui and was led by the 'six sense'. If Cook had chosen his target longitude more to the West, he might have overshot his aim. He might have never seen it. On the other hand, if he had chosen his target longitude more to the East, he would have run the risk of a prolonged tacking against the possible Westerly wind. As it happened, his choice was just right. The exceptional luck was with him and he sailed the remaining distance to the island with the following wind. These factors incline us to believe that Cook re-discovered Easter Island in a practical sense of discovering it. Therefore, the authors of the *Mémoire* to some extent had a right to emphasize the fact of this discovery.

Conclusions

We have approached the Forsters' *Mémoire* from two angles, apparently diametrically different: a social/literary one, and a navigational one. We wanted to throw light on similarities and differences between the French manuscript and other Forsterian works, showing the *Mémoire* as a valuable narrative piece. But we also desired to present it as an important work documenting a certain period or world exploration, and – in particular – obstacles waiting for the voyagers, obscuring their view or reality.

From our investigation we can infer that, in many circumstances, one needs not only technology and knowledge, but also intuition, political skills, and also luck, which people sometimes prefer to deem as coincidence. However, it would not be so manifest without a thorough reading of relevant documents – a reading that draws as well on humanities, as on sciences.

[38][G. Forster's note:] Juan Fernandez, properly so called or la de Tierra, and la Mas a fuera.

[39][G. Forster's note:] See the *Lives of the Governors of Batavia*. – It is there expressed 27° 04' S latitude, and 265° 42' E from Tenerife, or 110° 45' W from London.

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Wooden Figurines Collected by HMS Topaze on Easter Island

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Resumen

Estatuillas de madera recogidas por el HMS Topaze en la Isla de Pascua

El artículo identifica tres fotografías de los Archivos de la Congregación de los Corazones Sagrados de Jesús y María (Roma) tomadas a bordo de HMS Topaze, más probablemente en Valparaíso en diciembre 1868 por Paul-Emile Miot. Estas fotografías establecen la proveniencia de los cuatro artefactos: el hombre-pájaro Oc.1928,0517.1, colecciones del Museo Británico; moai pa'apa'a 6206, colecciones de Bernice P. Bishop Museum; el reimiro en forma de pez, la antigua colección Hooper, y un 'ao de la antigua colección Oldman.

Summary

This paper identifies three photographs from the archives of the Congregation of the Sacred Hearts of Jesus and Mary (Rome) as being taken onboard the HMS *Topaze*; most probably in Valparaiso (Chile) in December of 1868 by Paul-Emile Miot. These photographs established the provenience of the four artefacts: birdman Oc.1928,0517.1 from the collection of the British Museum; *moai pa'apa'a* 6206 from the collection of the Bernice P. Bishop Museum, the *reimiro* in the shape of a fish from the former Hooper collection, and an *'ao* from the former Oldman collection.

At all times, the sailors knew the necessity to make charts of the shores, marking the best anchoring places and depths. In the 18th century, the desire to share the newly-discovered landscapes and images of the people from faraway countries stimulated inclusion of the talented artists and painters into exploring crews; Cook's voyages are one of the best examples of such practice (Rüdiger and Bernard 1985–88). The works of the natives from far-away countries were also seen with fascination. The prospects of accurate documentation were much improved with invention of photography in 1822. However, the early photographic equipment was cumbersome and heavy; the low-sensitive photographic plates required long exposures and with heavy use of a tripod. Thus, the early photography was not much mobile, favoring studio-type pictures. In several decades, the technique was perfected enough to bring the cameras to remote places, including Easter Island (Thomson 1891). Yet there exist several earlier pictures documenting Rapa Nui artifacts; two of these depict the *moai* (statue) Hoa Hakananai'a standing onboard of a ship (Figure 1).

These photographs were published by Roussel (1926: 496–497), identifying the ship as HMS *Challenger*. The error of Roussel was 'possibly related to the diaries of Paymaster Routledge Richards of HMS *Challenger*. Although *Challenger* did not visit Easter Island, Richards included in his diary an account of the island 'culled from various authorities in the ship's library'(Van Tilburg 1992: 44). The Archives of the Congregation of the Sacred Hearts of Jesus and Mary (SS.CC.) include another remarkable picture Paq-5 (Figure. 2), with an inscription on the back *'Objets de l'Ile de Pâques à bord du «Challenger», navire de guerre anglais. Cap. G.R. Grant (1911). cf. «Herald», 23 août 1911. Photo reçue du P. Felix Jaffuel le 13-2-1926'*. Father Jaffuel most likely mixed up two *Challengers*: the ship famous for scientific trip around the world (1872–1876) and a more modern ship with the same name visiting Valparaiso in 1911. The fragment of lettering seen on the steering wheel (Figure 2) might have been the source of misunderstanding. However, under a closer look, the letters spell 'LENC' (as in Excellence) rather than 'LENG' (as

Since the 18th century Rapa Nui or Easter Island, as it was baptised by early European visitors, attracted many naturalists, archaeologists and anthropologists. It was only in 1984 when a series of the international conferences on 'Easter Island and the Pacific' was established. Three institutions, the German Archaeological Institute, the Christian Albrechts-University Kiel, and the HafenCity University Hamburg, united under the umbrella of the German Archaeological Expedition, organized the 9th International Conference in this series in the Ethnological Museum Berlin, Germany, which was held in June 2015. Researchers from 20 different countries presented their contributions of which 39 have been included in this publication. Their topics vary from archaeology, anthropology, history, linguistics, land-use and ecology, and heritage management to sociology. The papers cover the period from pre-contact to our times, with an emphasis on insular environments and resources across almost the entire Pacific.

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